



Tuka Curious Minds

MASTER ROBO WARS DESIGN

CONCEPT SUBMISSION

Robot Name: TURBO Ceros-T

Category: Robo Wars

Weight Classification: 5.kg Max Limit

Footprint Matrix: 420mm (Width) x 420mm (Length) x 55mm (Sleek Casing Height)

Inspiration Concepts discussed, sketched and AI generated to get to try generate a CAD design.

*Note the team are new to CAD, was a steep learning curve but tried to navigate through AI assistance, senior TCM members and YouTube tutorials

Version #1

Scrapped not feasible



We were thinking of doing a race car initially and reconsidered the material and components we had. We then thought there are 3 females and 1 male on the team with male coaches also and changed our mind to try out Robo Wars as females are under represented in this male dominated competition and career right?

Let's do this! Let's GO!

Game Play



Max damage to opponents??

Defence

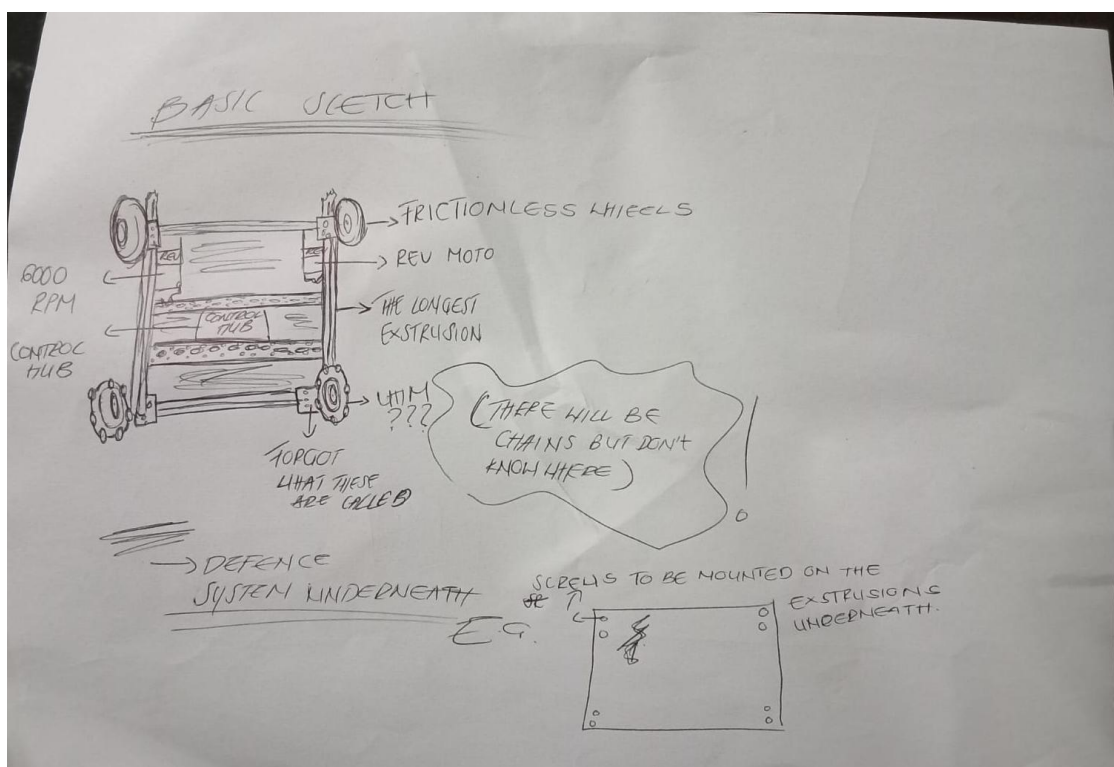
Speed – [running away!]

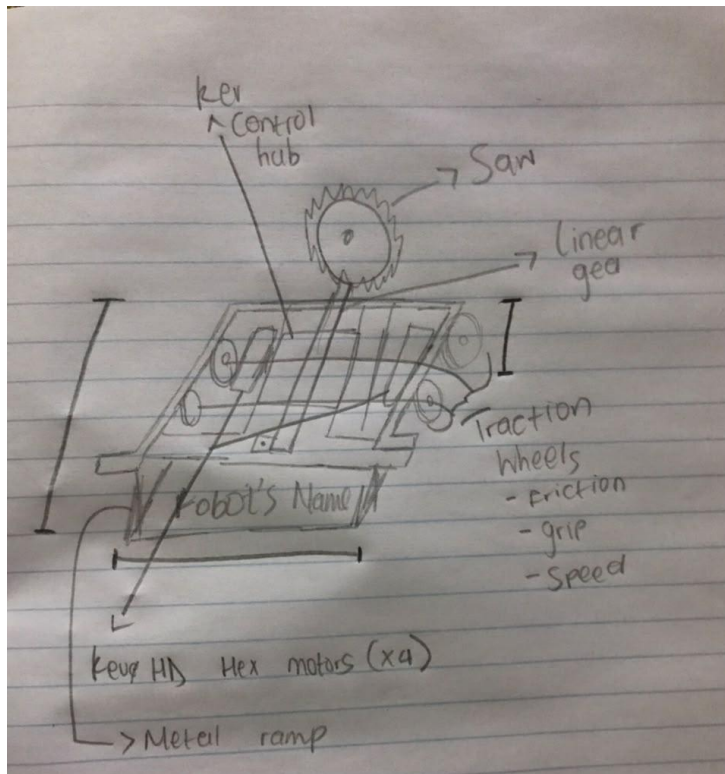
Last one standing...

Version 2

To consider – component parts such as traction wheels, extrusions vs c-channel, available motors.

Weapon system – 2 – 3 blades mounted on front of robot???





Version 3

Material - GoBuilder /Rev

Components. C channels that will hold onto the four traction wheels, because of their grip, speed and friction and then two on each side.

Motors - Single motor that drives both the wheels on each side.

- REV HTX motor, 6000 RPM, or

- GoBuilder 1:1 Yellow Jacket

Chain – Join that using a chain

Control - REV control hub

Weapons - solenoid that will stick out of the robot using a linear gear.

Front - A metal ramp that can pull, push, or lift up the opponent's robot and flip it over. motor, 6000 RPM, and we must make sure each motor has a good torque + speed for wheels

FINAL DESIGN THOUGHTS.....

Damage

Speed

FINAL CHOICE DESIGN

So we took all discussions and sketches and ran through AI to get a feel of what it could look like and compared all possible designs.

TCM

BUILT DIFFERENT.
BUILT TO WIN.

OVERVIEW
TCM is our answer to Robo Bumble. A massive vertical grinder blades up front to tear through anything in our way. Simple. Tough. Reliable.

BUILT BY
LEVEL BRAINED.

FRONT VIEW

50 cm

SIDE VIEW

50 cm

REAR VIEW

18 cm

TOP VIEW

50 cm

BUILT WITH REV + GOBILDA

REV ROBOTICS

GOBILDA

DIMENSIONS

50 cm

50 cm

18 cm (with Motor)

24 - 26 kg

WEAPON

- 3x 12" (30.5cm) SEPARATED GRINDER BLADES
- 1x REV 4000 STEEL BLADES
- Steel Drive with 175 lbs
- 90° Motors
- High inertia for maximum
- Power
- Clear Drive
- Adjustable Blade Spacing

BUILD DETAILS

- 1x 8" x 1/4" ALUMINUM SHEETS
- REV CHASSIS
- GOBILDA T20 SERIES
- GOBILDA 2000 20 RIES
- HARDWARE: STEEL, BRUNT WEDGE
- ALL HARDWARE COUNTERSINK

DRIVETRAIN

- 4x 500 SERIES GEAR MOTORS
- 1x 1/2" (12.7mm) WHEELS
- 4x 500 SERIES GEAR MOTORS
- 1x 1/2" (12.7mm) WHEELS
- 4x 500 SERIES GEAR MOTORS
- 1x 1/2" (12.7mm) WHEELS

ELECTRONICS

- REV ROBOT CONTROL HUB
- REV EXPANSION HUB
- 2x 12V 18AH BATTERIES
- POWER SWITCH + FUSE

REVENGE SAW

WEDGE + SAW RETALIATION ROBOT

CONCEPT

- Low weight front for ramming and pushing
- High torque 4WD for strong traction and stability
- Front-mounted spinning saw for retaliation and damage
- Low center of gravity for durability and control

TOP VIEW

120mm

40mm

SIDE VIEW

120mm

40mm

FRONT VIEW

120mm

40mm

Top Arm (Unpowered)

Saw Blade (12" / 30.5")

Saw Motor (REV 4000)

Wedge Front (Direct Hit)

Battery (12V 18AH)

Control Hub

Expansion Hub

Motor Controller

Drive Motors (4x REV 4000)

Wheels (12" Traction)

KEY FEATURES

- WEDGE FRONT**
Deflects hits and allows aggressive pushing.
- ACTIVE SAW**
High speed saw for retaliation.
- 4 WHEEL DRIVE**
Maximum traction, power and control.
- LOW PROFILE**
Harder to flip, better stability.

HARDWARE (REV / GOBILDA)

Drivetrain

- 4x REV 4000 New Motor
- 4x GOBILDA 20" Traction Wheel
- 2x REV 500 Series for Gearbox Data Motor Controller
- Adjustment Wrench (GOBILDA T20 Series)

Weapon (Saw)

- 1x REV 4000 New Motor
- 1x 12" / 30.5" Saw Blade (standard combat blade)
- Rev 4000 gear reduction (10T / 40T or 20T / 40T)
- Adjustment Wrench (GOBILDA T20 Series)

Electronics

- REV Control Hub
- REV Expansion Hub
- REV 12V 18AH Battery
- Power switch, wires, connectors

Other

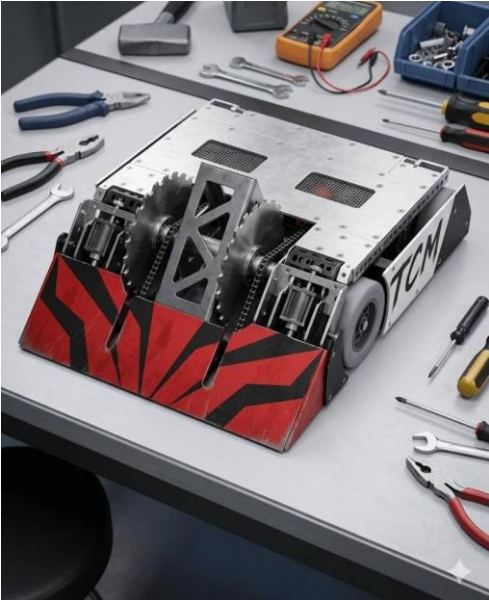
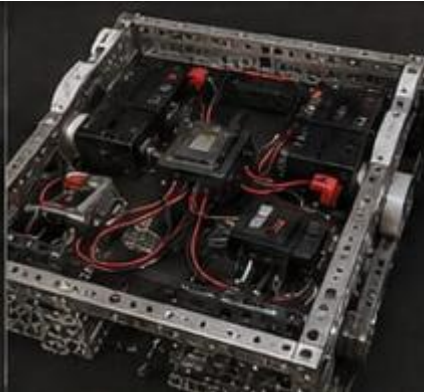
- Wedge made from 3mm Hardox / LHMW
- Saw Motor from 2000 Series
- Fasteners: standoffs, bearings, spacers

SOFTWARE

WIKWOK

Robot Simulator

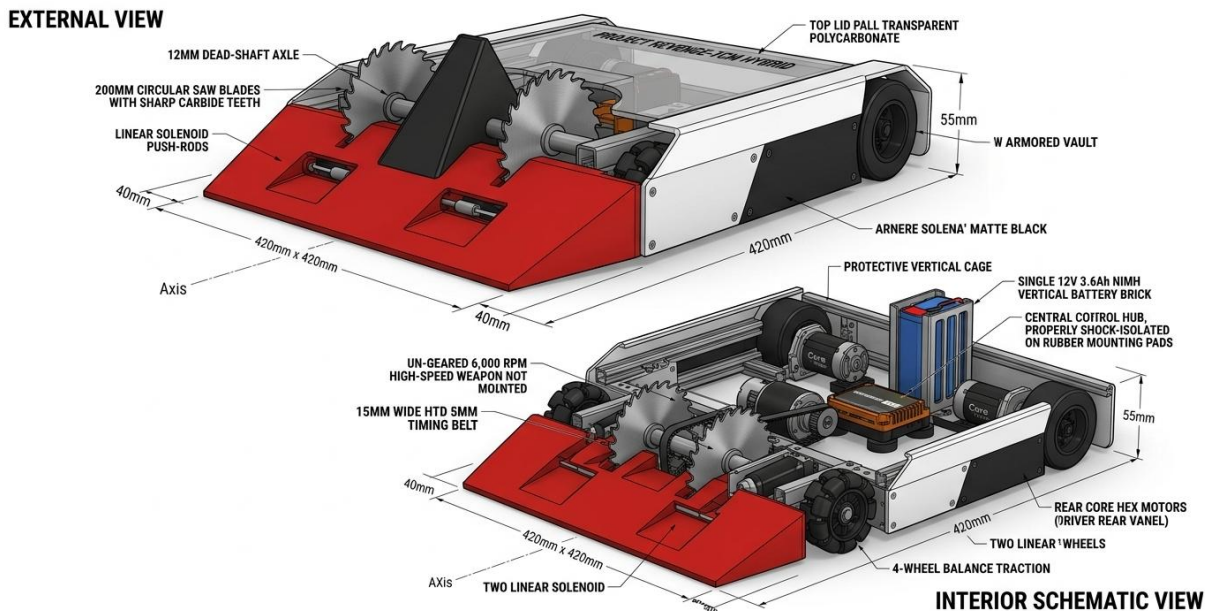
- Build your robot in Wikwok
- Test your robot in Wikwok
- Test your robot in Wikwok
- Test your robot in Wikwok





MEET TURBO CEROS-T!

Well what she'll possibly look like 😊

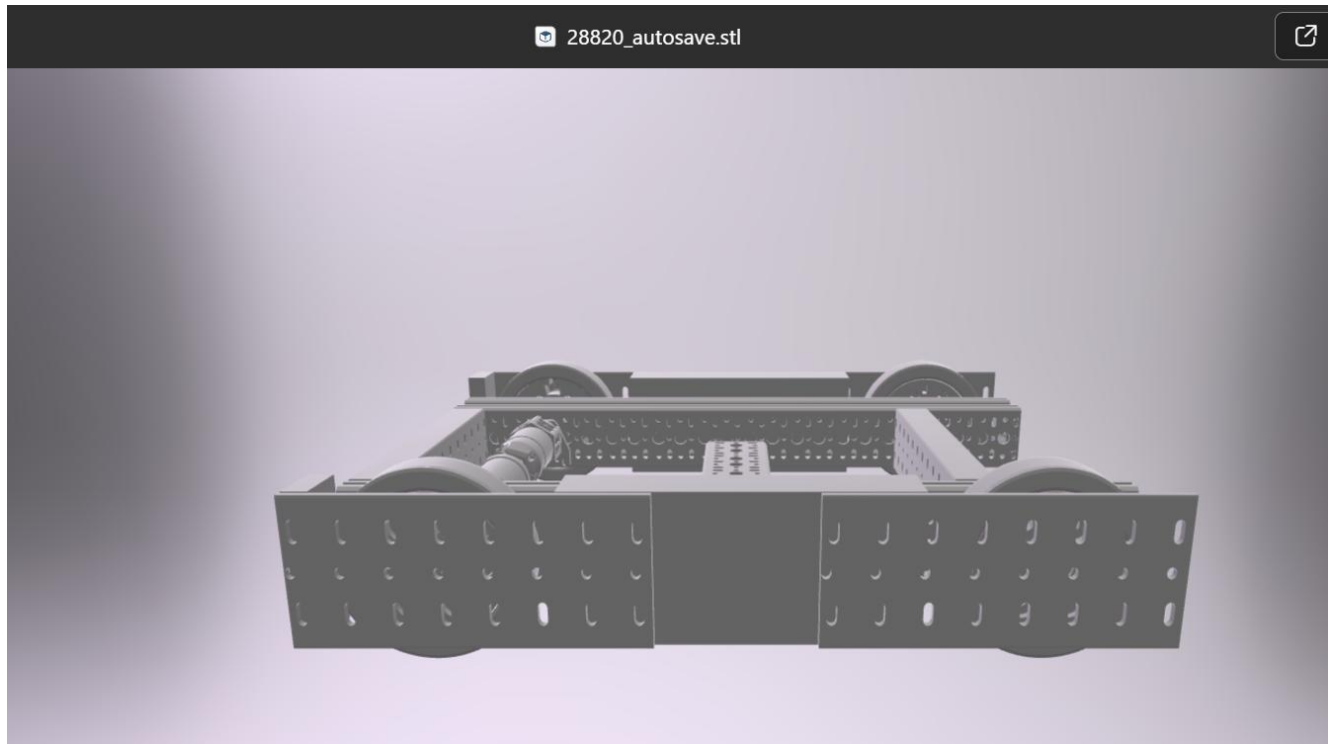


Final iteration [AI generated not 100% placement of blades, and components but this is the concept

Now for Blender/CAD attempt



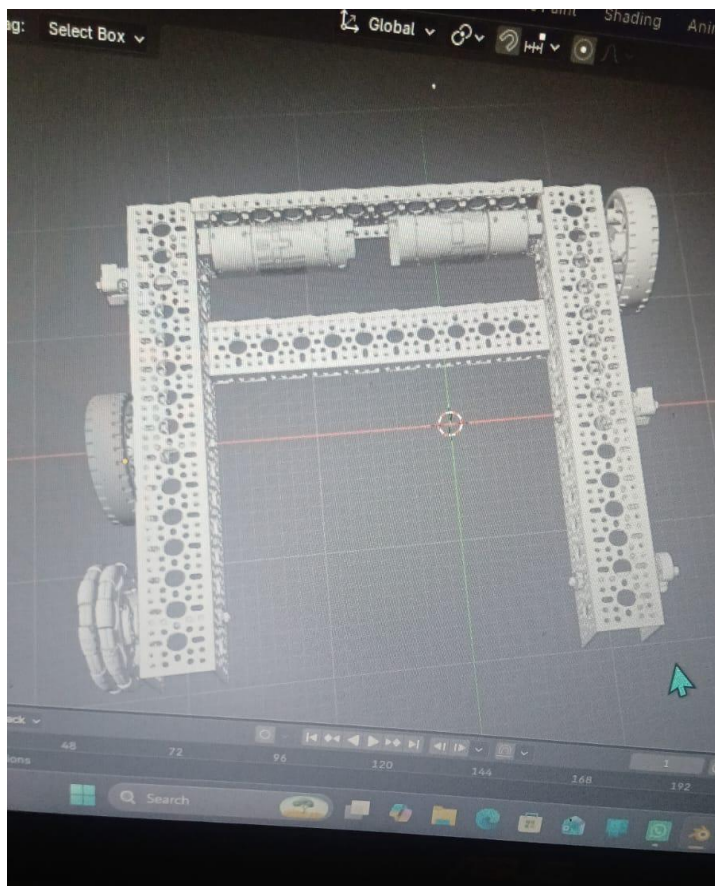
CAD #1

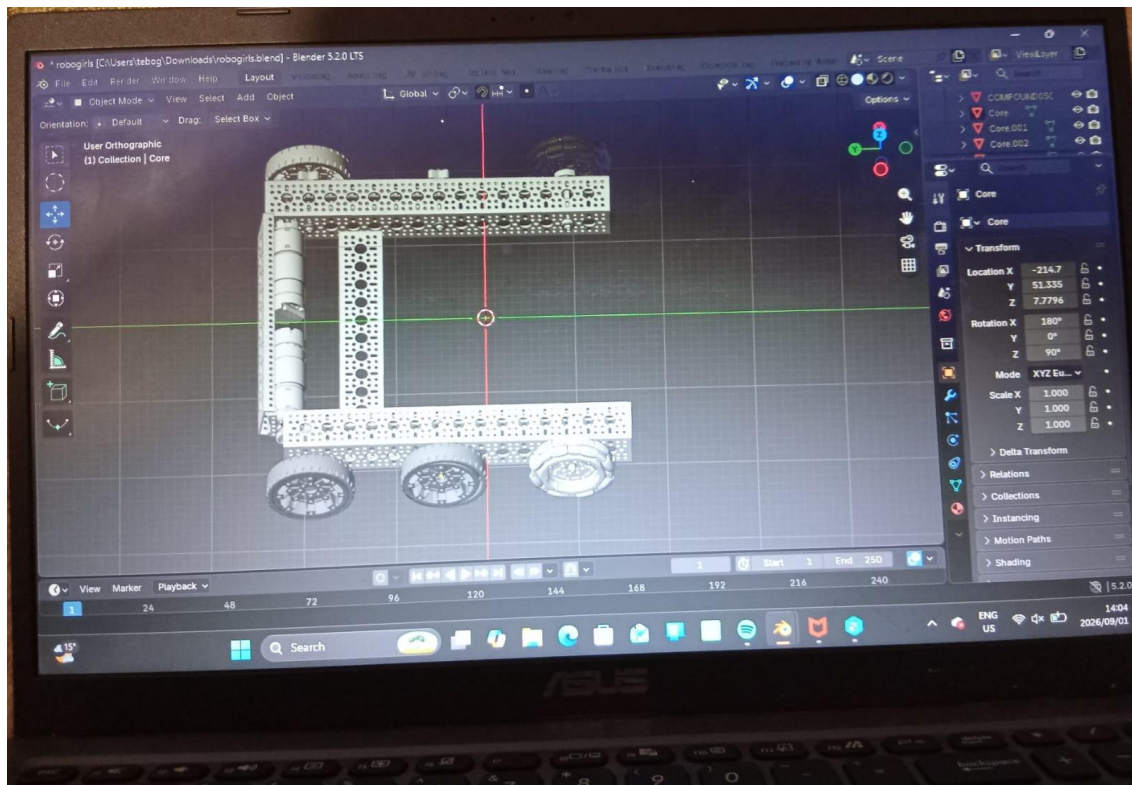


Thank you to senior coach Bonolo for getting us started and coaching session, but this is not easy as he makes it look!

We tried our best....

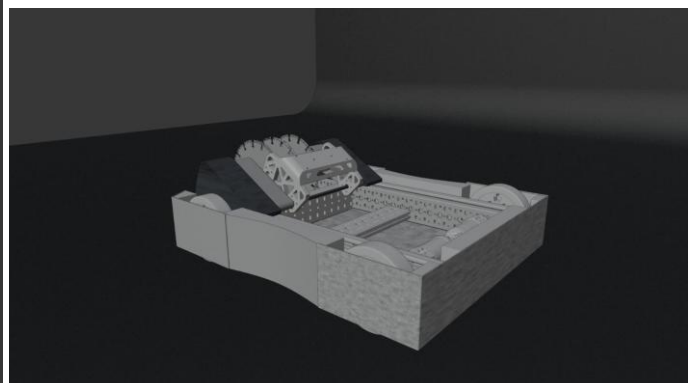
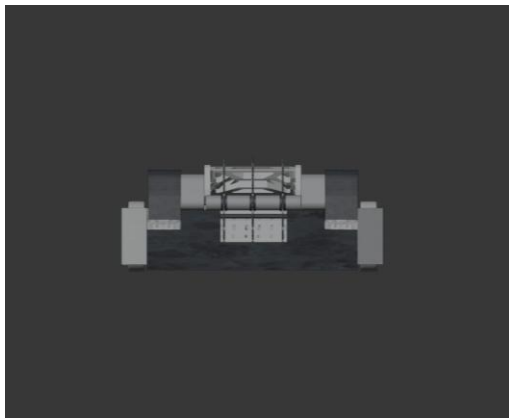
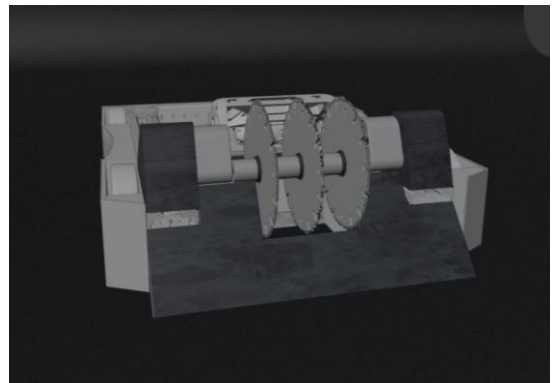
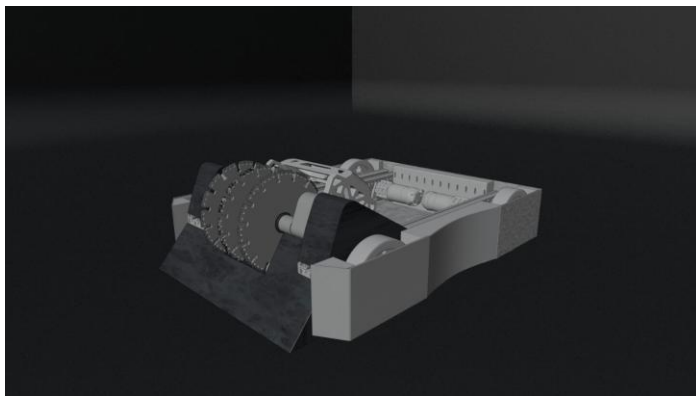
CAD #2

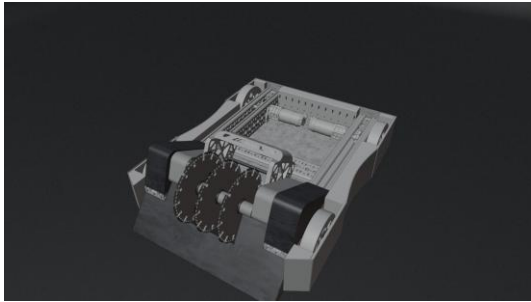
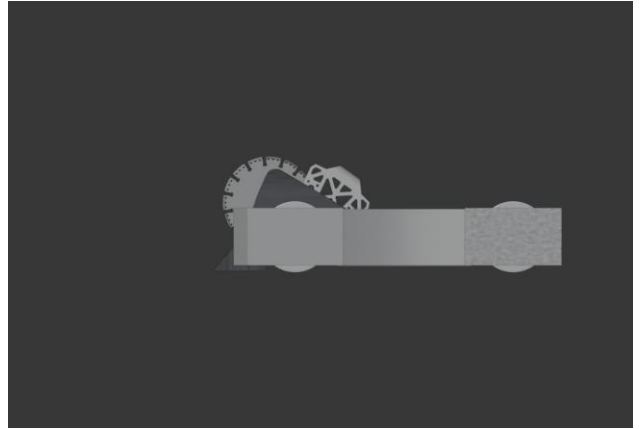
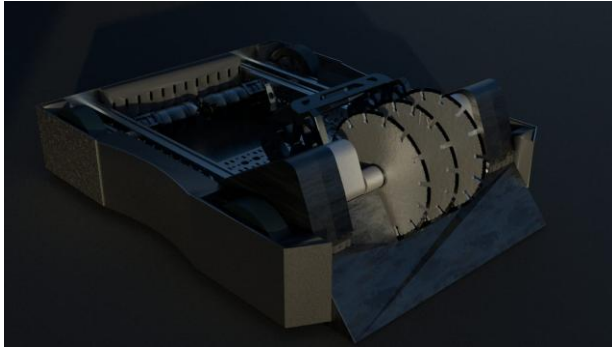




Not sure if the problem was the laptop, but the ASUS we using keeps stalling, crashing, restarting. This is how far we got

CAD #3





Thank you to Coach Bonolo for stepping in and assisting with final render version!

Should we get selected for next round, we hope to get a chance and opportunity to learn the basics and fundamentals of CAD as this was our first attempt as the girls squad trying to understand how to do this.

Thank you for considering TURBO Ceros-T to become one of the bots in the arena,

Making it through to the finals gives us a chance to compete against the boys!

Our team aspirations are to learn more of and gain exposure to programming using different languages, mechatronics and mechanics.

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